The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 NOV 93.

INCH-POUND

MIL-S-19500/504B(USAF)
10 August 1993
SUPERSEDING
MIL-S-19500/504A(USAF)
25 November 1991

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, DARLINGTON TRANSISTOR, NPN, SILICON, POWER
TYPES 2N6283 AND 2N6284, JANTX AND JANTXV

This specification is approved for use by the Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, Darlington, silicon, power transistors. Two levels of product assurance is provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

	P_T 1/		V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_J and T_{STG}	$R_{\theta JC}$
	T_C = +25°C	T_C = +100°C							
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N6283	175	87.5	80	80	7	20	0.5	-65 to +200	.857 max
2N6284	175	87.5	100	100	7	20	0.5	-65 to +200	.857 max

1/ Derate linearly at 1.17 W/°C above T_C > +25°C.

1.4 Primary electrical characteristics.

	h_{FE1} 1/	h_{FE2} 1/	$V_{CE(sat)1}$	$V_{CE(sat)2}$	$V_{BE(sat)1}$	Switching	
	V_{CE} = 3 V dc, I_C = 1 A dc	V_{CE} = 3 V dc, I_C = 10 A dc	I_C = 20 A dc, I_B = 200 mA dc	I_C = 10 A dc, I_B = 40 mA dc	I_C = 20 A dc, I_B = 200 mA dc	t_{on}	t_{off}
			V dc	V dc	V dc	µs	µs
Min	1,500	1,250					
Max		18,000	3.0	2.0	4.0	2	10

	C_{obo} V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	$\|h_{fe}\|$ V_{CE} = 3 V dc, I_C = 10 A dc, f = 1 MHz
	pF	
Min		8
Max	300	80

1/ Pulsed, see 4.5.1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 11, T-4A.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels
11	I_{CEX1} and h_{FE1}
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_{CEX1} = 100 percent of initial value or 100 µA dc, whichever is greater. Δh_{FE1} = ±40 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = +162.5°C ±12.5°C (see 4.5.2), $V_{CE} \geq$ 10 V dc.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	$V_{CE} \geq$ 10 V dc; ΔT_J between cycles ≥ +100°C; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on the devices shall be permitted.
B5	3131	$R_{\theta JC}$ = .857°C/W maximum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table I, group A, subgroup 2 herein.

4.4.3.1. Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
C6	1026	$V_{CE} \geq$ 10 V dc; ΔT_J between cycles ≥ +100°C. $t_{on} = t_{off}$ = 3 minutes minimum for 6,000 cycles. No heat sink or forced-air cooling on device shall be permitted.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector - emitter	3011	Bias condition D; I_C = 100 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N6283				80		V dc
2N6284				100		V dc
Collector - emitter cutoff current	3041	Bias condition D	I_{CEO}		1.0	mA dc
2N6283		V_{CE} = 40 V dc				
2N6284		V_{CE} = 50 V dc				
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = 7 V dc	I_{EBO}		2.5	mA dc
Collector - emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc	I_{CEX1}		0.5	mA dc
2N6283		V_{CE} = 80 V dc				
2N6284		V_{CE} = 100 V dc				
Base - emitter (nonsaturated)	3066	Test condition B; V_{CE} = 3 V dc; I_C = 10 A dc	V_{BE}		2.8	V dc
Base - emitter voltage (saturated)	3066	Test condition A; I_C = 20 A dc; I_B = 200 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)}$		4.0	V dc
Collector - emitter saturated voltage	3071	I_C = 20 A dc; I_B = 200 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		3.0	V dc
Collector - emitter saturated voltage	3071	I_C = 10 A dc; I_B = 40 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		2.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE1}	1,500		
Forward-current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE2}	1,250	18,000	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Forward-current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 20 A dc; pulsed (see 4.5.1)	h_{FE3}	500		
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector - emitter cutoff current 2N6283 2N6284	3041	Bias condition A; V_{BE} = 1.5 V dc; V_{CE} = 80 V dc V_{CE} = 100 V dc	I_{CEX2}		5.0	mA dc
Collector - emitter saturated voltage	3071	I_C = 10 A dc; I_B = 40 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)3}$		2.0	V dc
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 3 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE4}	200		
Subgroup 4						
Pulse response						
Turn-on time		(See figure 1); V_{CC} = 30 V dc; I_C = 10 A dc; I_B = 40 mA dc	t_{on}		2.0	µs
Turn-off time		(See figure 1); V_{CC} = 30 V dc; I_C = 10 A dc; I_{B1} = I_{B2} = 40 mA dc	t_{off}		10	µs
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 3 V dc; I_C = 10 A dc; f = 1.0 MHz	$\|h_{fe}\|$	8	80	
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 3 V dc; I_C = 10 A dc; f = 1 kHz	h_{fe}	700		
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		300	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area continuous (dc)	3051	T_C = +25°C; t = 1s; 1 cycle; (see figure 2)				
Test 1 (Both device types)		V_{CE} = 8.75 V dc; I_C = 20 A dc				
Test 2 (Both device types)		V_{CE} = 30 V dc; I_C = 5.8 A dc				
Test 3 2N6283		V_{CE} = 80 V dc; I_C = 100 mA dc				
2N6284		V_{CE} = 100 V dc; I_C = 100 mA dc				
Safe operating area (switching)	3053	Load condition C; (unclamped inductive load); (see figure 3) T_A = +25°C; $R_s \leq 0.1\ \Omega$; $t_r + t_f \leq 15$ ns; duty cycle ≤ 2 percent				
Test 1		t_p = 80 µs; (vary to obtain I_C); $R_{BB1} \geq 50\ \Omega$; $V_{BB1} \geq 10$ V dc; $R_{BB2} = \infty$; V_{BB2} = 0; I_C = 20 A dc; $V_{CC} \geq 50$ V dc; The coil used shall provide a minimum inductance of 1 mH at 20 A. (For reference only; two coils in parallel (Super Electric Corporation type S16884 or equivalent).)				
Test 2		t_p = 1 ms; (vary to obtain I_C); $R_{BB1} \geq 50\ \Omega$; $V_{BB1} \geq 10$ V dc; $R_{BB2} = \infty$; V_{BB2} = 0; I_C = 500 mA dc; $V_{CC} \geq 50$ V dc; The coil used shall provide a minimum inductance of 100 mH at 500 mA. (For reference only; two coils in series, 80 mH and 20 mH windings (Super Electric Corporation type S16884 or equivalent).)				

See footnote at end of table.

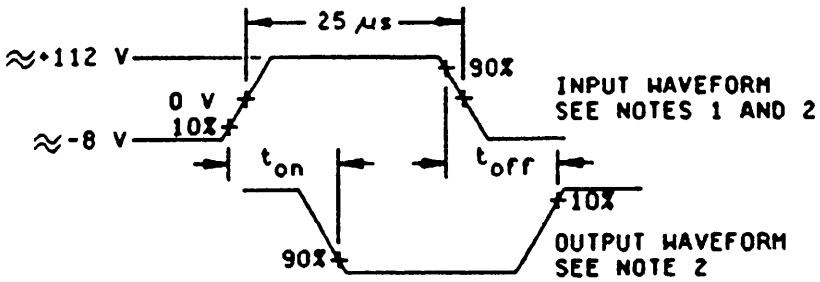
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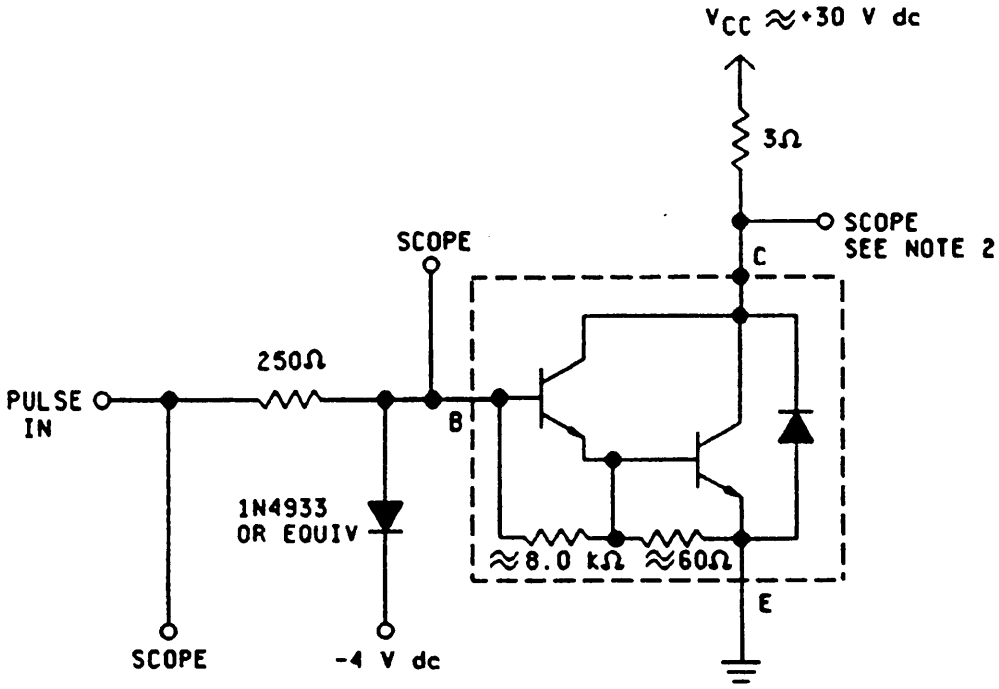
TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued.						
Safe operating area (switching)	3053	Load condition B; (clamped inductive load); T_A = +25°C; R_s = 0.1 Ω; $t_r + t_f \leq 1.0$ µs; duty cycle ≤ 2 percent; t_p = 1 ms (vary to obtain I_C); R_{BB1} = 50 Ω; V_{BB1} = 10 V dc; R_{BB2} = 100 Ω; V_{BB2} = 1.5 V dc; V_{CC} = 25 V dc; I_C = 20 A dc; $R_L \leq 2$ Ω; L = 5 mH; (Four coils in parallel, 20 mH windings) (Triad C-48u or equivalent).				
2N6283		Clamp voltage = 80 +0, -5 V dc				
2N6284		Clamp voltage = 100 +0, -5 V dc				
		Device fails if clamp voltage not reached				
End point electrical measurements		See table II, steps 1 and 3				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics:

   $t_r \leq 20$ ns, $t_f \leq 20$ ns, $Z_{out} = 50\ \Omega$, PW = 25 µs, duty cycle ≤ 2 percent.

2. Output waveforms are monitored on an oscilloscope with the following characteristics:

   $t_r \leq 2$ ns, $Z_{in} \geq 20$ kΩ, $C_{in} \leq 11.5$ pF.

3. Resistors shall be noninductive types.
4. The dc power supplies may require additional by-passing in order to minimize ringing.

FIGURE 1. Pulse response test circuit.

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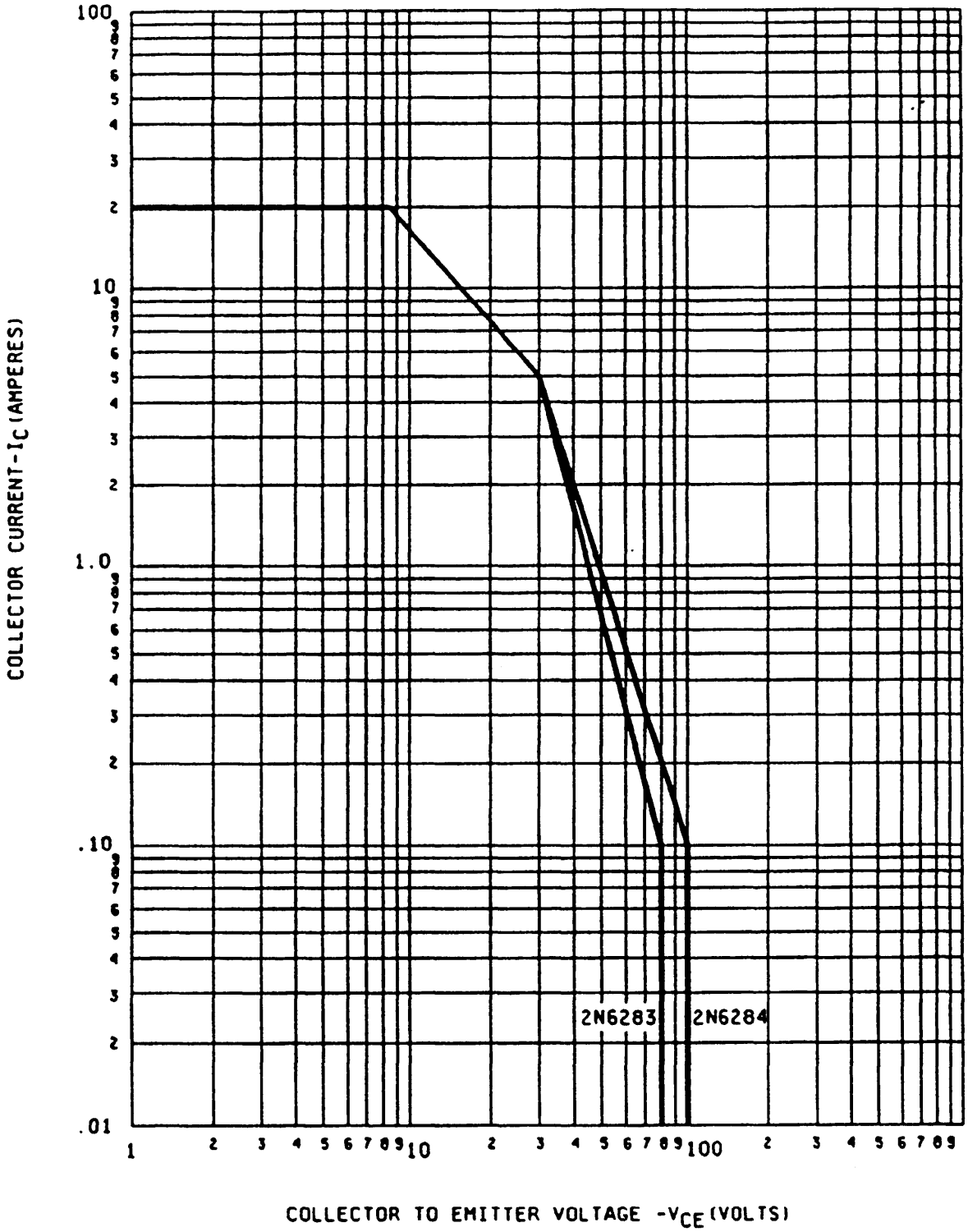


FIGURE 2. Maximum safe operating graph (continuous dc).

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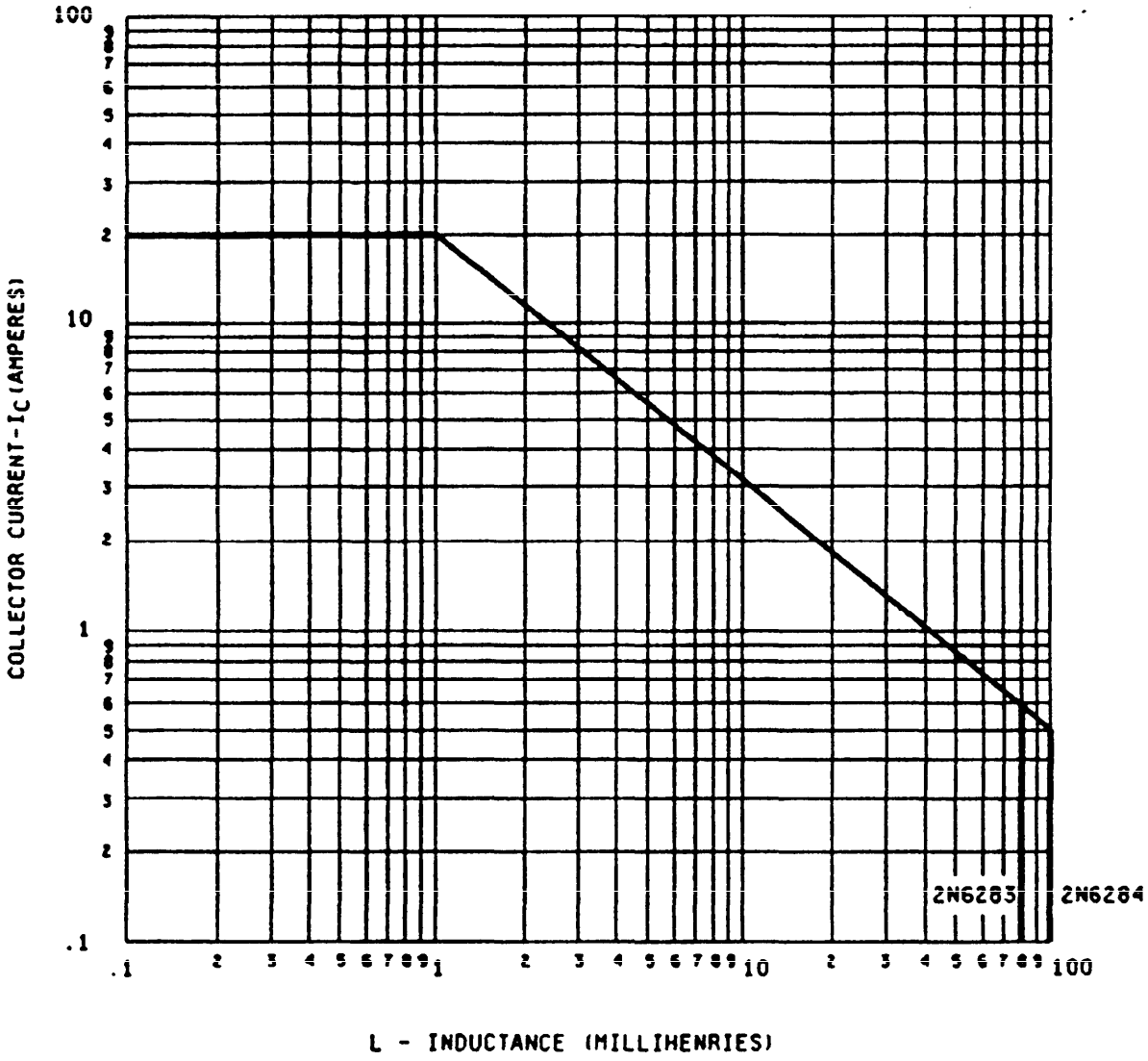


FIGURE 3. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead formation and finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 17

Review activities:
Air Force - 13, 19, 85, 99
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/504B(USAF)	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DARLINGTON TRANSISTOR, NPN, SILICON, POWER, TYPES 2N6283 AND 2N6284, JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
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